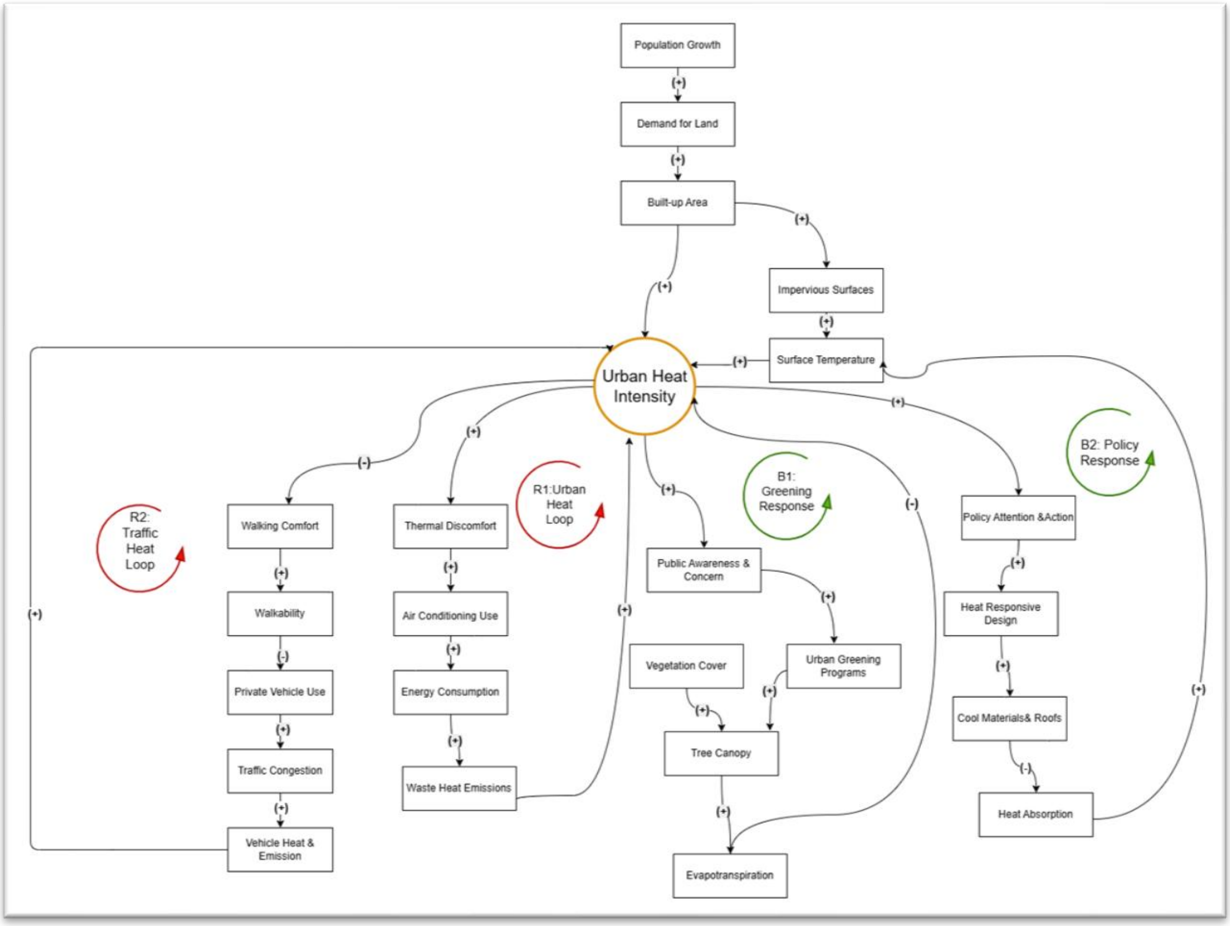
Causal Loop Analysis of Urban Heat in Dehiwala

Study Area: Dehiwala Junction

Identified Problem: Urban Heat Intensity

1. Causal Loop Diagram (CLD)



2. Loop Analysis

The Causal Loop Diagram helps explain how different factors interact and contribute to urban heat in Dehiwala. It also shows how some processes worsen the problem while others help reduce it.

<u>Reinforcing Loop (R1): Heat Trap Loop</u>

When Urban Heat Intensity increases, people experience greater Thermal Discomfort. To Stay comfortable, they use more Air Conditioning (AC). Increased AC use leads to higher

Energy Consumption, which results in more Wate Heat Emissions being released into the surrounding environment. This additional heat further increases Urban Heat Intensity.

- ❖ As a result, the cycle continuous and heat problem becomes worse over time.

Why is it a Reinforcing Loop?

This loop strengthens itself. An increase in urban heat leads to actions that generate even more heat. Since the effect keeps growing in the same direction, it is identified as a Reinforcing Loop (R1).

<u>Balancing Loop (B1): Greening Response Loop</u>

As Urban Heat Intensity increases, people become more aware of the problem and express greater concern. This encourages the implementation of Urban Greening Programs such as tree planting and green space development. These actions increase the Tree Canopy, which improves Evapotranspiration and provides natural cooling. As a result, Urban Heat Intensity decreases.

- ❖ This process helps reduce the heat problem.

Why is it a Balancing Loop?

This loop works against to the increase in urban heat. The cooling effect created by trees and vegetation helps bring the system back toward a more comfortable condition. Therefore, it is classified as a Balancing Loop (B1).

3. Leverage Point Identification

Leverage points are places within a system where a small change can create a significant positive impact.

Leverage Point 1: Impervious Surfaces

Impervious surfaces such as concrete roads, pavements, and buildings absorb and store large amounts of heat. In a highly urbanized area like Dehiwala, increasing impervious surfaces directly raise surface temperatures and contributes to urban heat.

Reducing these surfaces or replacing them with more heat-friendly materials can significantly lower temperature levels.

Leverage Point 2: Tree Canopy

Tree canopy plays a key role in reducing urban heat. Trees provide shade and support evapotranspiration, both of which help cool the surrounding environment.

Increasing tree cover strengthens the natural cooling process and supports the balancing loop that works against rising urban temperatures.

4. Planning Intervention

Urban Tree Planting and Shaded Streets Program Explanation

One of the main reasons for increasing urban heat in Dehiwala is the lack of vegetation in highly built-up areas. Many roads, commercial areas, and public spaces have limited tree cover, which increases exposure to direct sunlight and raises surface temperatures.

To address this issue, the local authority can implement an Urban Tree Planting and Shaded Streets Program. This program would focus on planting shade trees along major roads, around public spaces, schools, and commercial areas. Existing green spaces should also be protected and maintained to prevent further loss of vegetation.

We expect increasing tree canopy cover will provide shade and improve evapotranspiration, helping to reduce surface temperatures and urban heat intensity. A cooler environment can improve outdoor comfort, encourage walking, and reduce the dependence on air conditioning.

This intervention directly strengthens the Greening Response Loop (B1) identified in the CLD and helps counteract the reinforcing effects of the Heat Trap Loop (R1). Over time, it can contribute to a more comfortable and sustainable urban environment in Dehiwala.

5. Conclusion

Urban heat intensity is a growing issue in Dehiwala due to increasing built-up areas, impervious surfaces, and reduced vegetation cover. The Causal Loop Diagram shows that some factors, such as air conditioning use and traffic emissions, can worsen the problem, while urban greening and policy actions can help reduce it. Therefore, increasing tree canopy and promoting greener urban spaces are important steps towards creating a cooler, healthier, and more sustainable environment in Dehiwala.

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